PHARMACOKINETICS STUDY IN RAT

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LIST OF ABBREVIATIONS

ABBREVIATION	DESCRIPTION
%	Percentage
°C	Degree Celsius
ADME	Absorption distribution metabolism & excretion
AH	Animal house
BQL	Below quantification limit
CD	Caesarean derived
CO_2	Carbon dioxide
DMPK	Drug metabolism & pharmacokinetics
HPLC	High performance liquid chromatography
hr	Hour
i.v	Intravenous
ID	Identification
IPM	Intellectual property management
K_2-EDTA	Di-potassium ethylene di amine tetra-acetic acid
L/kg	Litre/ kilogram
LC-MS	Liquid chromatography and mass spectrometry
LLOQ	Lower limit of quantification
LNB	Lab note book
mg/kg	Milligram/ kilogram
min	Minute
mL/kg	Millilitre/ kilogram
mL/min/kg	Millilitre/ minute/ kilogram
Mol. Wt.	Molecular weight
MS	Microsoft
Na_2 EDTA	Sodium ethylene di amine tetra-acetic acid
NaCl	Sodium chloride
NCA	Non-compartmental analysis
NR	Not Reported
O_2	Oxygen
p.o	Per oral
PD	Pharmaceutical development
PK	Pharmacokinetics
PK-PD	Pharmacokinetics- pharmacodynamics
PS	Pure solution
PPE	Personnel protective equipment

ABBREVIATION	DESCRIPTION
RPM	Revolutions Per Minute
SAR	Structure activity relationship
SM	Spiked matrix
SM PEG	Spiked matrix with Polyethylene Glycol - 200 vehicle
SOP	Standard operating procedure
TK	Toxicokinetic
ULOQ	Upper limit of quantification

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1. OBJECTIVE

The objective of study is to determine the pharmacokinetic parameters of test article (s) in rat after dosing via various routes of administration like intravenous, per oral, subcutaneous, intramuscular and intraperitoneal etc. Biological samples (blood, urine, feces and tissues etc.) will be collected from rats to estimate and calculate the test article concentration and pharmacokinetic parameters respectively.

2. SCOPE

The procedure described in this protocol will be used to determine the concentration and pharmacokinetic parameters of a test article (s) in rat blood, urine, feces and tissues after intended route of administration at Aurigene Pharmaceutical Services Ltd, India.

3. RESPONSIBILITY

All personnel performing this study are responsible to follow the process outlined in this protocol, at Aurigene Pharmaceutical Services Ltd, India.

4. INTRODUCTION

The Pharmacokinetics (PK) study is to evaluate the absorption, distribution, metabolism and excretion (ADME) profile of test articles. One of the primary reasons cited for the failure of new drug candidates in clinical trials is the lack of a suitable pharmacokinetics (PK) profile. Despite that, valuable insight is obtained from *in vitro* ADME screening assays; *in vivo* drug exposure is still emphasized by drug discovery teams when making decisions about molecules within a Structure Activity Relationship (SAR). Plasma concentration versus time profiles is essential for deriving a more detailed analysis of drug exposure in terms of both PK and toxicokinetic (TK). Pharmacokinetics profile obtained from preclinical species will helps in predicting human pharmacokinetic parameters following different scaling tools (such as allometry scaling, PK/PD tools) which helps in designing clinical trial studies.

5. SAFETY AND PRECAUTIONS

During the conduct of the study, the study personnel should take necessary safety precautions and use personal protective equipment (PPE) such as facemasks, gloves, lab coat, goggles, shoes etc.

DMSO or other organic solvents:

- Keep away from sources of ignition.
- Do not breathe vapour or mist.
- Avoid contact with skin, eyes and clothing.

6. MATERIALS AND EQUIPMENT

6.1 EQUIPMENT/INSTRUMENTS

Equipment	Make
Analytical balance /Micro balance	Sartorius/ Denver/ TB-215D
Centrifuge with Plate / Tube rotors	Eppendorf & Sigma/ Kubota
Deep Freezer (-80°C)	Sanyo VIP series/ Thermo

Equipment	Make
Heating pad	Classic electronics
HPLC	Shimadzu/ Waters
LC-MS/MS	AB Sciex/Waters Xevo
Gas Anesthesia System	Orchid scientific
Nitrogen evaporator	Turbovap®LV
pH meter	Sartorious/ PB-11/ Lab India
Pipettes	Eppendorf/ Gilson, Finn pipette/ Brand
Probe sonicator	Ultrasonic Processor / 04714-5, Antylia Scientific
Refrigerator	LG
Sonicator	Ultrasonic
Surgicals	Braintree scientific/ Kent Scientific/ Asian Surgical
Vortexer	IKA vortex genius-3//Heidolph/Spinix
Water bath sonicator	Ultrasonic LC-30, Elma/ Life care
Wound clips	Braintree scientific

Note: Equivalent equipment/ instruments shall be used based on the availability.

6.2 CHEMICALS/CONSUMABLES

Chemicals	Make; Catalogue No.
Adhesive	Fevikwik
CO_2	Vinayaka gas agency/ Praxair
Ethanol	Changshu Yangyuan Chemical
Heparin	Celon laboratory and Troika
Isoflurane	Abbott
K_2-EDTA	Sigma-Aldrich, ED2P
Ketoprofen	Ranbaxy
Methylcellulose	Sigma-Aldrich
Micro tips	Tarsons
N- Methyl -2 Pyrrolidone (NMP)	Sigma-Aldrich
O_2	Vinayaka gas agency/ Praxair

Oral gavage	Braintree/ Harvard
Polyethylene Glycol	Sigma-Aldrich
Povidone-Iodine	Cipla
Silicone catheter	Instech Solomon/Scientific Commodities, INC.
Sodium chloride	Sigma-Aldrich, 71380
Surgical suturing thread	Lifeline B.B Silk Suture-USP
Syringe(1 mL, 2 mL, 5 mL)	BD discardit-II syringe
Tween 80	Sigma-Aldrich

Note:

- All the chemicals used for conducting study shall be documented in the LNB.
- Equivalent grade chemicals/ reagents shall be used on need basis.
- Suitable anesthesia and consumables shall be used as per IAEC approved/ literature specified, if above mentioned anesthetic and consumables or materials are not available.

7. STUDY DETAILS

7.1 TEST ARTICLE DETAILS

Test substance id	
Mol. Wt.	
Salt Wt.	
Salt correction factor	
Batch no.	
Purity (%)	
Storage conditions	

7.2 TEST SYSTEM

7.1.1 JUSTIFICATION FOR SELECTION OF TEST SYSTEM

Rats are a commonly used rodent species in the evaluation of pharmacokinetics and Toxicokinetics of new chemical entities accepted by the regulatory authorities.

7.1.2 ANIMALS

The following information applies:

Species : Rat
Strain : Crl:CD (SD)
Source : In-house or external (as per the specification)
Sex : Male/Female
Age : 7-10 weeks

Body Weight : 220-320 g
Number : 3 animals/ dose
Animals will be housed three per cage, in polypropylene cages containing irradiated corncob bedding material along with paper cuttings.

7.1.3 ANIMAL MANAGEMENT

Temperature	20-24°C
Relative humidity	30-70%
Air changes	10-15 times per hour
Lighting	12 hr day light and 12 hr darkness
Caging	Three in transparent cages placed in rolling racks.
Identity	Animals will be identified by unique tail marking.
Acclimatization	On receipt from the animal house/ suppliers, the animals will be examined for external signs of ill health prior to acceptance. The rats will then be acclimatized over a period of at least 3 to 7 days prior to utilization.
Diet	TEKLAD certified global 14% protein rodent diet (Harlan Laboratories, Inc.)
Water	Chlorinated Reverse Osmosis (RO) treated water will be provided

Commented [RR1]: Acclimatization were not following can we remove this point?

8. EXPERIMENTAL PROCEDURE

8.1 PROCEDURE FOR DOUBLE JUGULAR VEIN CANNULATION IN RATS

8.1.1 MATERIAL PREPARATION FOR SURGERY

Rodent surgeries should be performed under aseptic conditions (aseptic techniques, gloves, caps, masks and sterile instruments etc.) as well as in separate surgical and animal preparatory areas.

All instruments used in surgeries shall be ensured for the following.

- Steam sterilized prior to surgery
- kept on sterile non-porous drapes during use
- Shall be cleaned of blood and debris by brushing or wiping with sterile saline and sterile gauze sponges between surgeries.
- If contamination has occurred, instruments must be placed in a chemical agent or a glass bead sterilizer for an appropriate period of time for the method to be effective (or else, the pack must be replaced by a new one).
- If a chemical agent is used, instruments must be rinsed with sterile water or saline before being used for the next animal.

Surgical gloves and blades should be changed between surgeries for each animal.

During the entire surgery procedure, the incisions and surgical areas must be kept moist with a disinfectant solution.

Cannula preparation:

Cut a piece of silicon tube (0.51 mm I.D x 0.94 mm O.D) approximately 10 cm in length. Fix a retention bead made from the scalp vein set onto the cannula with the help of an adhesive (fevikwik). Allow it to dry for 30 min at room temperature and then dry it in an oven at 60˚C for 10 min or ready to use catheters can be procured from authorized vendors.

8.1.2 SURGICAL PROCEDURE

Anesthetization: The anaesthesia shall be initiated by placing the animal into a hermetic plexiglas box with isoflurane 3-5% evaporated in medical grade oxygen (flow rate of 2 mL/min).

After anaesthesia induction, the animal is prepared by shave the hair from the back of neck, thorax, the junction of thorax and right forelimb with a trimmer, and then transferred to the surgery place.

Thereafter, the anaesthesia is maintained using a face mask with Isoflurane 2-3% evaporated in medical grade oxygen (flow rate of 0.6-1 mL/min).

During the entire surgical session, the anaesthesia is carefully monitored by the experimenter and the Isoflurane concentrations and the oxygen flow rates are adapted accordingly.

In House Surgery: Make small incisions at the back of the neck (2-3 mm) and at the jugular implantation region (5-7 mm, middle of the throat).

Place the rat on its back on the thermal bed (temperature maintained at 37°C) with nose pointing towards the surgeon. Tape the forelimbs with the paper tape and subcutaneously guide the tunneling needle from the jugular incision to the neck incision.

Guide each catheter through the tunneling needle and then pull it out via the neck wound.

The catheters must now be in place with their tips close to the jugular incision and their ends exteriorized at the neck incision.

Connect the catheter ends to a syringe containing 0.9% NaCl.

Carefully separate and isolate the right jugular vein by removing the tissue around it.

Insert a glucose strip/ pair of forceps under the vein for stabilizing the vein and having a rigid surface for effective catheterization. Close permanently the vein with knots of the silk ligatures immediately after the bifurcation.

After filling that portion of vein with blood, tie the anterior ligature tightly to occlude blood flow returning to the heart. Tie the posterior ligature with a single loose half-hitch which can be used to create tension on the vessel during venotomy and catheter insertion. A venotomy must be made posterior to the anterior ligature using iris scissors.

Make a fine incision into the vein, about 1/3 to 1/2 of its total diameter, at 1-2 mm above anterior ligature.

Insert the catheter tip into the vein incision in the direction of the heart.

Release temporarily the tension from the posterior ligature to allow the complete insertion of the small catheter part into the vein.

Adjust the catheter length to 2.5 cm.

Check catheter placement by aspirating some blood and gently flush with 40 to 100 µL of 0.9% NaCl.

Close the anterior and posterior ligature using two to three knots each with moderate strength.

Repeat the same procedure for implantation of a catheter into the left jugular vein. Fix both the catheters properly in the back of the neck

After double implantation and functionality test of the catheters, close the surgery wound by wound clips or stitches.

Stop the isoflurane anesthesia.

Apply the povidon-iodine solution to the sutured area and cannulated animals will be individually caged.

Analgesics should be given pre-emptively, so before incisions are made the plasma drug level has reached the effective concentrations. Depending upon the drug (e.g.,Ketoprofen 5 mg/kg SC; Buprenorphine 0.01 mg/kg SC; Meloxicam 1-2 mg/kg SC, etc), usually one hour before surgery.

Transfer the animal to an individual cage for recovery. Provide paper material and a red color plastic tunnel for nesting and occupation in their cages. Ensure free access to water and food.

During recovery, visually monitor the general condition of the rats and its consumption of food and water.

Check catheters again for patency and replace the saline with 0.05 mL lock solution (heparin (500 units/ mL): glycerol: saline in 1:5:4 ratio).

Lock solution should be replaced with 0.9% NaCl before start of experiment.

The recommended recovery period is 4 days (96 hrs) after surgery.

8.1.3 BODY WEIGHT

Weigh the rats prior to dose administration and record the weight in raw data file and/ or LNB.

9. DOSING

9.1 PREPARATION OF DOSING SOLUTION FOR INTRAVENOUS ADMINISTRATION

The formulation shall be prepared by using 10% NMP, 90% PEG-200 for intravenous dosing. The formulation shall be prepared as per the following procedure.

Add corresponding amount of NMP to each of the vials containing pre-weighed amount of the compound and vortex vials till a clear solution is obtained, if required apply ultrasonication and water bath (up to 50°C max) then add corresponding amount of PEG-200.

For cassette studies, prepare solutions of each compound individually in NMP. Pipette equivolume of each compound into a new vial and add corresponding amount of PEG-200.

For cassette dosing, the compounds which do not yield a clear solution in NMP must be excluded and an equivolume of NMP should be added to the cocktail.

9.2 INTRAVENOUS ADMINISTRATION (FED CONDITION)

Administer the test article as a bolus injection (20 to 30 sec) through the left jugular cannula at an administration volume of 0.5 mL/ kg body weight.

The exact start time of dosing and exact dose volume shall be documented in LNB.

9.3 PREPARATION OF DOSING SUSPENSION FOR ORAL ADMINISTRATION

The formulation shall be prepared using 0.1% Tween 80, 0.5% MC in water for oral dosing.

Preparation of formulation:

Add corresponding amount of formulation premix into the vial containing pre-weighed amount of the compound and vortex till a homogenous suspension (no floating particles) and no more powder is sticking

to the walls of the vial, if required apply ultrasonication and water bath (up to 50°C max) to get a homogenous suspension.

For cassette studies, add the formulation premix individually to vials containing each of the compounds. Mix equivolumes of each compound into a new vial to obtain a homogenous suspension. The compounds which do not yield a good suspension must be excluded and an equivolume of vehicle should be added to the cocktail

9.4 ORAL ADMINISTRATION (FED CONDITION)

Administer the test article as a suspension into the esophagus using a gavage needle at an administration volume of 5 mL/ kg body weight.

The exact start time of dosing and exact dose volume shall be documented in LNB.

9.5 GROUPING, ROUTE OF ADMINISTRATION AND TARGET DOSES

For crossover design, the animals should be first dosed by intravenous route with blood sampling at designated time points. Following the 24 hours' sample, the oral dose shall be administered.

Discrete Study Design (Crossover)

Group	No. of Rats	Rat Identification	Route of administration	Dose (mg/kg)
I	3	1-3	Intravenous	0.3 or 1
I		1-3	Oral	3 or 10

IV & PO Discrete Study Design (single arm studies)

Group	No. of Rats	Rat Identification	Route of administration	Dose (mg/kg)
I	3	1-3	Intravenous	0.3 or 1
II	3	4-6	Oral	3 or 10

Cassette Dosing Study Design (Crossover)

Group	No. of Rats	Rat Identification	Route of administration	Dose (mg/kg)
I	3	1-3	Intravenous	0.3 or 1
I		1-3	Oral	3 or 10

Cassette Dosing Study Design (single arm studies)

Group	No. of Rats	Rat Identification	Route of administration	Dose (mg/kg)
I	3	1-3	Intravenous	0.3 or 1
II	3	4-6	Oral	3 or 10

Note:

- For cassette dosing 5 to 6 compounds or as per study plan shall be used and the specific information such as dose, concentration and dose volume shall be documented in study sheet or LNB.
- Rats shall be dosed in FED condition.

10. CLINICAL SIGNS

Animals should be checked before and following dosing and record the clinical signs observed, if any.

11. SAMPLING

Blood sample collection method:

About 20 µL volume of normal saline in the canula is drawn and discarded. About 15-20 µL of fresh blood is collected from the jugular vein catheter (without EDTA) and then transferred into a microcentrifuge tube. ~20-30µL normal saline is infused into the canula.
Rat blood, immediately pipet/discharge 10 µL (exact volume) of fresh blood into 20 µL of EDTA solution* in a 0.5 or 1.5 mL microcentrifuge tube. A vortex step can be done after transferring blood into EDTA solution.

Sampling time points following post i.v. dose: 0.083, 0.25, 0.5, 1, 2, 4, 7 and 24 hr.
Sampling time points following post p.o dose: 0.25, 0.5, 1, 2, 4, 7 and 24 hr
All the time points (samples) should be collect within 5% of their nominals.
Exact dosing and sampling times should be recorded. After the last sampling time rats should be given for euthanization.

The blood samples (containing 10 µL blood + 20 µL EDTA solution) after collection shall be stored to -20°C as soon as possible or placed directly on dry ice until analysis.

Plasma sample collection method

a. Collect approximately 60 µL of fresh whole blood from jugular vein catheter and then pipet/transfer fresh blood into an EDTA-coated tube.
b. Transfer ~35 µL of blood for Hematocrit measurement.
c. Prepare now the plasma before dilute with Blood/EDTA solution(1:2;v:v) solution: Centrifugate the remaining volume in the tube @4°C to get ~12 µL EDTA plasma. Then take out accurately 10 µL plasma and add 20 µL blank blood-EDTA solution 1:2 (10 µL blood + 20 µL EDTA solution)
d. For other time points with no need for hematocrit measurement, approximately 25 µL of fresh whole blood should be taken at Step .a and continue with step b and c

The EDTA blood sample tube can be maintained on wet ice (+4°C) for a short period of time only. Maximum 20 minutes after blood sampling, centrifuge the EDTA whole blood at 6000 x g, 4°C for 10 min. plasma, immediately pipet/discharge 10 µL (exact volume) of plasma into 20 µL of Blood/EDTA solution(1:2;v:v) in a 0.5 or 1.5 mL microcentrifuge tube *. A vortex step can be done after transferring plasma into (Blood/EDTA solution(1:2;v:v)) and samples stored to -20°C as soon as possible or placed directly on dry ice until analysis.
Note: The samples shall be maintained at + 4°C by keeping on wet ice during collection for a short period of time only: the maximum time from sample collection to being stored in -20°C should be no more than 30 minutes.

12. HANDLING OF DEVIATIONS OF IN LIFE DATA

- When the deviation in dose and dose volume is greater than 1.5 folds or less than 0.5 fold of theoretical value, the deviation should be flagged in the report; Pharon table should be updated to report the actual value(s)

- When the actual collection time is beyond the acceptable time range, the deviation should be highlighted in the report; Pharon table should be updated to report the actual value(s)

Scheduled Collection Time	Acceptable Time Range
0 – 15 minutes	± 1 minute
16 – 30 minutes	± 2 minutes
31 – 45 minutes	± 3 minutes
46 – 60 minutes	± 4 minutes
61 minutes – 2 hours	± 5 minutes
2 hours 1 minute – 8 hours	±10 minutes
8 hours 1 minute – 24 hours	± 20 minutes
24 hours 1 minute onwards	± 60 minutes

- When the measured hematocrit value is outside the range (min/max), should re-measure the hematocrit within 24h post-dosing. In case the re-measured hematocrit is still out of range, should check (historical data of animal, etc) and flag it in the report both results, initial and re-measures if valid, in the hematocrit table.
- Note: a re-measurement of hematocrit is possible and independent of study type.

Hematocrit	Literature value		Range
	Min	Max	
%	%		%
Mouse	45	36	54
Rat	46	37	55
Rabbit	36	29	43
Dog	42	34	50
Monkey	41	33	49
Minipig	38	30	46

13. METHOD & PHARMACOKINETIC ANALYSIS OF BLOOD SAMPLES

Analyze the concentration of the test article in blood samples following appropriate method by using LC-MS/MS.

Tune the compound to get optimized parameters of test article to analyze followed by chromatographic conditions.

Assess the matrix (ion suppression) effect and recovery at one mid-range concentration (e.g. 50 ng/ mL) level by injecting 3 different samples:

- Compound spiked in pure solution.
- Compound spiked into blank matrix extract and 0.7% (v/v) blank PEG-200).
- The calibration standard (50 ng/ mL).

13.1 CALIBRATION & UNKNOWN SAMPLES

Calibration samples preparation:

Transfer 27 µL blank blood-EDTA mix ((10 µL blank blood + 20 µL EDTA solution)) into pre-labeled micro centrifuge tubes and spike 3 µL test compound solutions of appropriate concentration to prepare the calibration standards in the range of 0.5 to 5000 ng/ mL or 0.1 to 5000 ng/ mL, depending on the sensitivity of the compound. The Calibration standards must be prepared in duplicates including Standard zero (Blank +ISTD, C00).

Samples preparation:

Aliquot 30 µL unknown sample (10 µL blood, pre-pipette into 20 µL EDTA solution)/Calibration samples/Quality control samples into pre-labeled micro centrifuge tubes, crash with 100 µL acetonitrile containing internal standard (carbamazepine/ verapamil/ telmisartan/ warfarin/ tolbutamide/ celecoxib).

Vortex the samples for about 5 min and centrifuge at 13000 rpm for 5 min at 4°C.

Transfer 80 µL supernatant into pre-labelled micro centrifuge tubes, add 80 µL water and vortex.

Note: Nominal CC concentrations were calculated only at 10 µL final blood volume. (Dilution excluded). However, the actual dilution was 3 µL cc solution + 27 µL blank blood-EDTA (1:2 v/v) and total volume 30 µL. Below Annexure illustrate the preparation of the Working stock and Calibration Standards.

13.2 PREPARATION OF SM, SM PEG

SM: Pipette 100 µL of blank blood + EDTA mix (1:2 v/v) into a pre-labelled micro centrifuge tube and vortex for approximately 1 minute.

SMPEG.: Pipette 116 µL blank blood-EDTA mix (1:2 v/v) and 4 µL 0.7% aqueous PEG solution into a pre-labelled micro centrifuge tube and vortex for approximately 1 minute. Final concentration in blood: 0.07% PEG 200. (To prepare 0.7% aqueous PEG solution, dispense 0.7 µL PEG 200 and make up the volume to 100 µL with water).

Sample Preparation: Aliquote 30 µL of SM and SM PEG, each in triplicates and crash with 100 µL acetonitrile containing internal standard (carbamazepine/ verapamil/ telmisartan/ warfarin/ tolbutamide/ celecoxib).

Vortex the samples for about 5 min and centrifuge at 13000 rpm for 5 min at 4°C.

Transfer 80 µL supernatant into 1.5 mL micro-centrifuge tube and add 80 µL water, add 3 µL calibration standard solution (50 ng/ mL).

13.3 PREPARATION OF PURE SOLUTION

Aliquote 30 µL of acetonitrile/water (1/1) into a pre-labelled micro centrifuge tube and add 100 µL acetonitrile containing IS.

Vortex the samples for about 5 min and centrifuge at 13000 rpm for 5 min at 4°C. Transfer 80 µL supernatant into microcentrifuge tube and add 80 µL water and 3 µL calibration standard solution (50 ng/ mL).

13.4 REANALYSIS

In-case of instrument malfunction, communication failure the samples shall be reanalyzed/reinjection, along with QC samples. If the QCs meets acceptance criteria, then the sample concentrations shall be considered. If the QCs not meets acceptance, then the sample data cannot be used. If possible, samples

shall be re-extracted along with calibration standards and QC's. If not possible, then appropriate notes shall be added to the BA remarks, describing what has occurred, and listing the impacted samples, that have no data. The same note shall be added to the CT Pharon table for those samples, in the Comm Samp column.

13.5 ACCEPTANCE CRITERIA OF BIOANALYSIS

Calibration standards range should cover at least 0.5 to 5000 ng/ mL with a minimum of 7 calibration levels or 0.1 to 5000 ng/ mL with a minimum of 9 calibration levels in this range and a blank. Below annexure contains detail procedure for preparation of working stock and calibration standard.

Calibration standards acceptance criteria:

An LLOQ of 0.1 or 0.5 ng/ mL is expected to be achieved regularly (> 75% of the studies).

75% of standards should be within +/- 30% of nominal value, including LLOQ and ULOQ.

QC acceptance criteria:

Run 3 QCs (low, mid, high), in duplicate or 6 QCs as singlets (can be re-injection of calibration stds). 67% of QCs as singlets must be within+/- 30% of nominal value.

Internal standard acceptance criteria:

The peak area of the internal standard(s) should be monitored and evaluated for each analytical run. The peak area of the internal standard for the study specimens should be 50% to 150% of the overall mean peak area of internal standard for the entire run. (i.e. Cs/STDs, QCs, zeros).

If there are IS peak area outliers (IS peak area < 50% or > 150% of the overall flat and linear mean of the entire run), Re-inject or test alternate IS, adjust chromatography. If the new result is within assay acceptance, it will be noted that internal standard tracked with this sample, will report the new value, with the valid internal standard.

If it is not possible to obtain internal standard within specification, then the samples impacted will be listed in the Analytics tab, BA remarks comments, and those values will not be reported to the Pharon concentration time table, but will be included in the raw data Export tab of the report. The cells for concentration values in Pharon table should be empty. A note of "data not used, IS failed" should be made in the "CommSamp" column of Pharon table.

Blank interference and carryover impact

The area of the pre injected blank peak area must be ≤ 30% of the peak area of the subsequent sample, standard or QC.

Carryover effects should be monitored (for example peak area of upper standard, compared to subsequent blank) to ensure carryover effects do not significantly impact the accuracy or precision of the results.

In case of carryover during bioanalytical run, reinject CS00 between 2500ng/mL calibration standard and before first study sample and check that the area of the last CS00 is ≤ 30% of the LLOQ.

14. DATA RECORDING

The data shall be tabulated in MS excel based pharmacokinetic study template provided by Novartis and the final data shall be uploaded in to NIBR NEXT and the final data should be recorded in Aurigene services LNB for internal reference.

15. REPORTING

Data should be reported in an MS excel format, based on the need, MS word based reports shall be be prepared.

16. DATA ARCHIVAL

All the raw data, chromatograms and reports should be stored in the server at Aurigene Pharmaceutical Services Limited, India for a period of five years or as specified by sponsor. Assay details and analytical procedures should be captured in LNB and the same should be archived as per the instructions/ procedure of Intellectual Property Management (IPM) department.

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17. Annexure

DMSO stock preparation

Weighed compound was dissolved in calculated quantity of DMSO to obtain concentration of 1 mg/mL.

Preparation of working stock and calibration stock for 0.1 ng/mL LLOQ

Serial dilution of DMSO stock was done with Acetonitrile: Water (1:1) to obtain working standard and calibration standard as given in below table.

Stock Solution ID	Concentration of stock (µg/mL)	Volume of stock (µL)	Diluent volume(µL)	Final volume(µL)	Working Stock Concentration (µg/mL)	Working stock ID
Example	1000.000000	25.00	1475.00	1500	16.666667	WS-11
WS-11	16.666667	250	250.00	500	8.333333	WS-10
WS-10	8.333333	250	250.00	500	4.166667	WS-9
WS-11	4.166667	100	400.00	500	0.833333	WS-8
WS-8	0.833333	100	400.00	500	0.166667	WS-7
WS-7	0.166667	100	400.00	500	0.033333	WS-6
WS-6	0.033333	100	400.00	500	0.006667	WS-5
WS-5	0.006667	250	250.00	500	0.003333	WS-4
WS-4	0.003333	250	250.00	500	0.001667	WS-3
WS-3	0.001667	250	250.00	500	0.000833	WS-2
WS-2	0.000833	200	300.00	500	0.000333	WS-1

Matrix Spiked Calibration Curve Standards						
Working stock ID	Working Stock Concentration (µg/mL)	Working Stock Volume (µL)	Blank blood-EDTA mix Volume (µL)	Blood volume (µL)	Final Blood Matrix Conc. ng/mL	Calibration Standard ID
WS-11	16.666667	3	27	10	5000.00	C011
WS-10	8.333333	3	27	10	2500.00	C010
WS-11	4.166667	3	27	10	1250.00	C009
WS-8	0.833333	3	27	10	250.00	C008
WS-7	0.166667	3	27	10	50.00	C007
WS-6	0.033333	3	27	10	10.00	C006
WS-5	0.006667	3	27	10	2.00	C005
WS-4	0.003333	3	27	10	1.00	C004
WS-3	0.001667	3	27	10	0.50	C003
WS-2	0.000833	3	27	10	0.25	C002
WS-1	0.000333	3	27	10	0.10	C001

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Preparation of working stock and calibration stock for 0.5 ng/mL LLOQ

Serial dilution of DMSO stock was done with Acetonitrile: Water (1:1) to obtain working standard and calibration standard as given in below table.

Stock Solution ID	Concentration of stock (ng/mL)	Volume of stock (µL)	Diluent volume(µL)	Final volume(µL)	Working Stock Concentration (ng/mL)	Working stock ID
Example	1000000.00	25.00	1475	1500	16666.67	WS-9
WS-9	16666.67	250	250	500	8333.33	WS-8
WS-8	8333.33	250	250	500	4166.67	WS-7
WS-7	4166.67	100	400	500	833.33	WS-6
WS-6	833.33	100	400	500	166.67	WS-5
WS-5	166.67	100	400	500	33.33	WS-4
WS-4	33.33	100	400	500	6.67	WS-3
WS-3	6.67	250	250	500	3.33	WS-2
WS-2	3.33	250	250	500	1.67	WS-1

Matrix Spiked Calibration Curve Standards						
Working stock ID	Working Stock Concentration (ng/mL)	Working Stock Volume (µL)	Blank blood-EDTA mix Volume (µL)	Blood volume (µL)	Final Blood Matrix Conc. ng/mL	Calibration Standard ID
WS-9	16666.67	3	27	10	5000.00	C09
WS-8	8333.33	3	27	10	2500.00	C08
WS-7	4166.67	3	27	10	1250.00	C07
WS-6	833.33	3	27	10	250.00	C06
WS-5	166.67	3	27	10	50.00	C05
WS-4	33.33	3	27	10	10.00	C04
WS-3	6.67	3	27	10	2.00	C03
WS-2	3.33	3	27	10	1.00	C02
WS-1	1.67	3	27	10	0.50	C01

18. REFERENCES

- Novartis Guidelines
- APHIS (1997): policy no #11 Animal Care Resource Guide.
- Pharmaceutical Research, Volume 10, No. 07 1993, Brian Davies and Tim Morris.
- www.ahc.umn.edu/rar/safety.html#Surveillance
- SOP GLP-AF-0013; environmental enrichment
- APS-DMPK-712: Bioanalytical Method Development and Qualification for PK studies in Discovery Phase
- IAEC protocols

19. CHANGE AND AMENDMENT HISTORY

Change History		
Version	**Changes made/ Reason for changes**	**Effective date**
3.0	1. Change in sample processing procedure as per new Novartis Discovery BA protocols for Aurigene 2. Deletion of procedure for sampling of formulation as formulation analysis not performed in LCMS. 3. Infrared light replaced with thermal pad during surgical procedure.	10 OCT 2022
4.0	1. Included preparation of DMSO stock, Working Standard and Calibration Standard.	22 May 2023

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